



FROM PROMPTS TO

LOOP ENGINEERING

The workflow shift in AI-native development.

prompt -> context -> harness -> loop

Every two years the place where your effort actually matters climbs one step higher.

Swipe ->

THE REFRAME

You're not getting better at prompting. The skill didn't improve -- it moved up the stack.



THE REFRAME

You're not getting better at prompting. The level you work at is **moving up the stack.**

Prompt -> context -> rig -> loop: each step automates the craft below it and pushes your effort up a level.

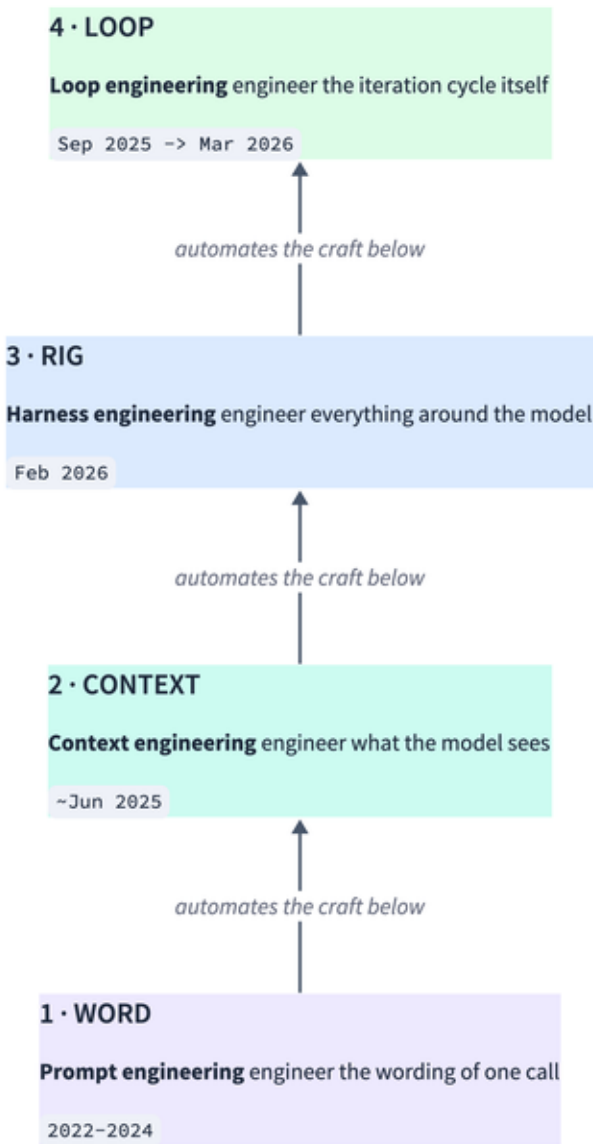
From Prompts to Loop Engineering -- the four-era arc

THE STAIRCASE

Prompt -> context -> harness -> loop. The unit of work you own rises one step every two years.

The staircase: four eras, one moving target

Each step automates the craft below it and pushes your effort up a level.

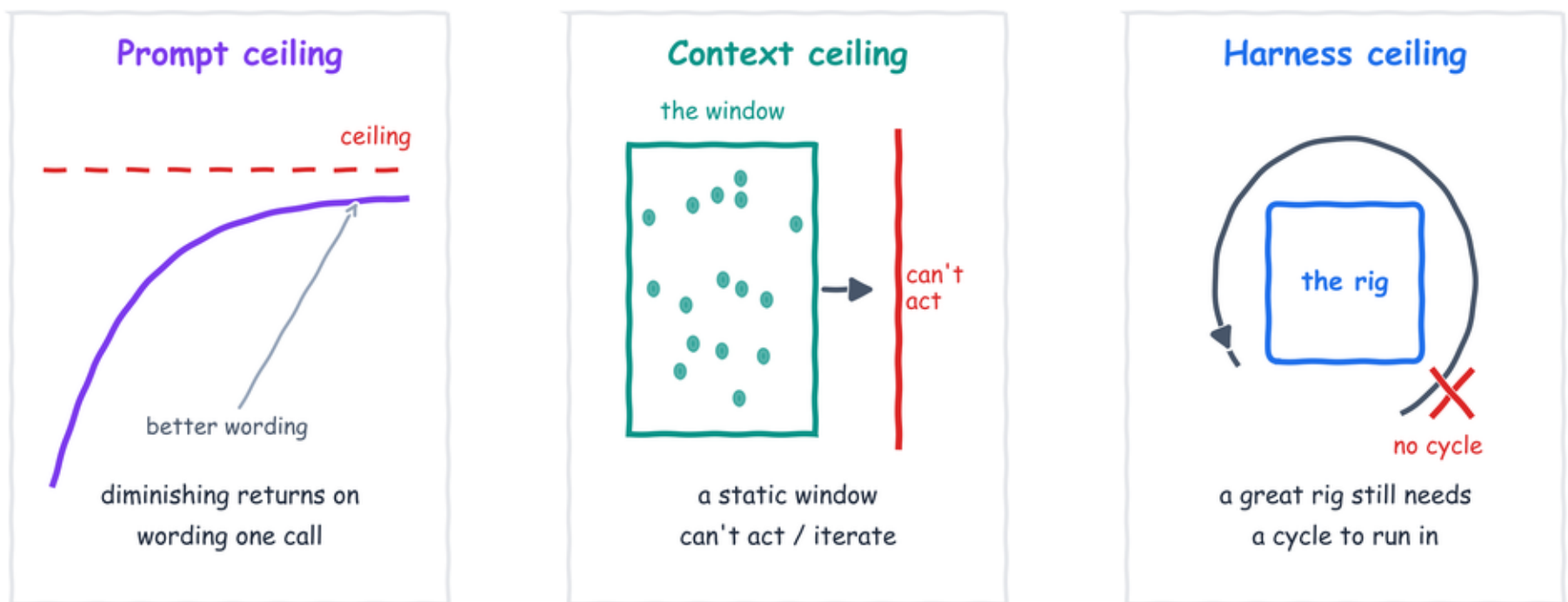


Sources: trend-research; Fowler 'Exploring Gen AI' index; Bockeler (Feb 2026); Willison (Sep 2025).

ERAS 1-3 HIT A WALL

Rewording hits diminishing returns, a curated context window still can't act, and a 100-line rig has no cycle to run in.

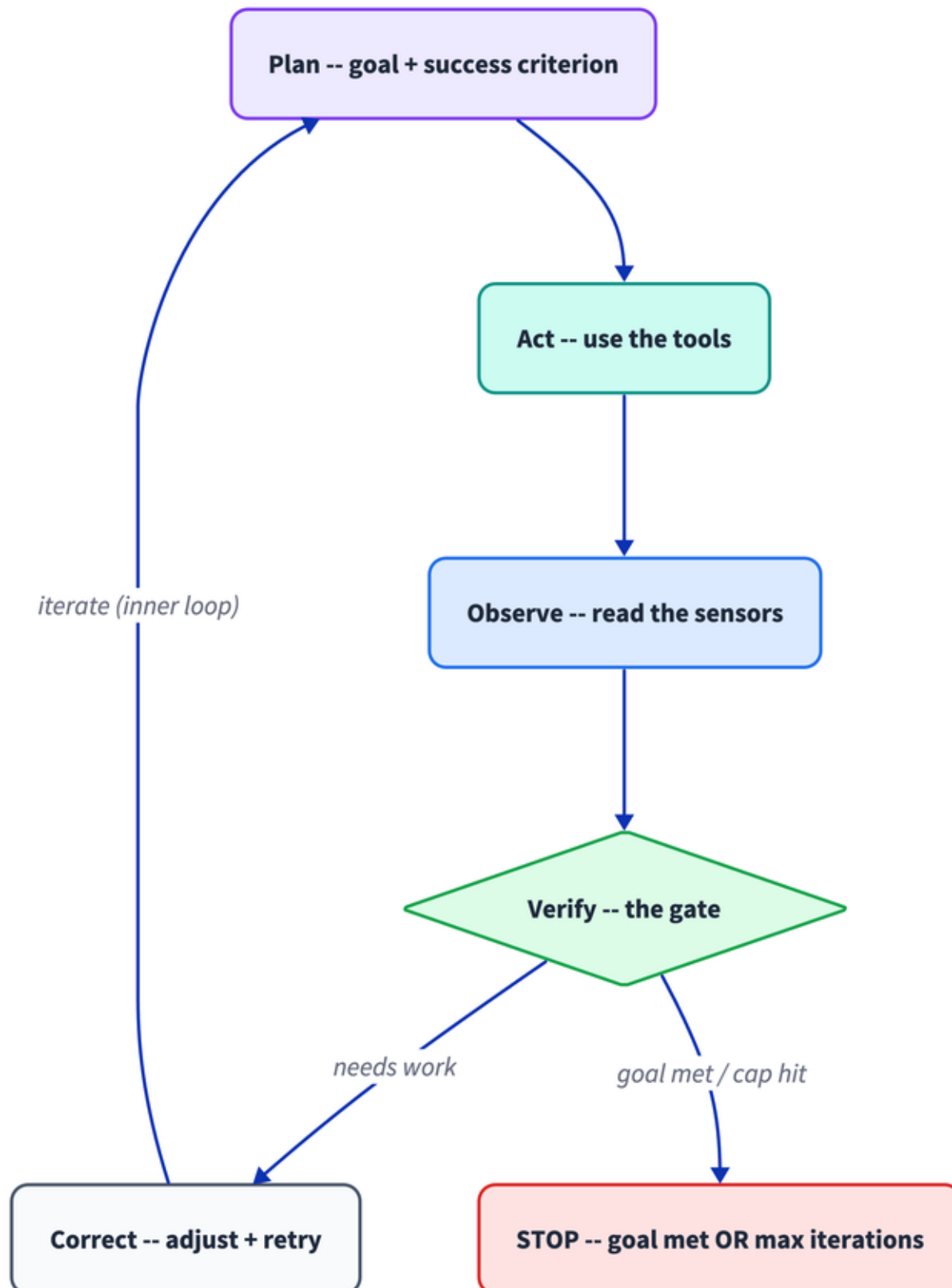
Why each era hit a ceiling and pushed you up a level



Proof tie-in: a minimal harness (mini-SWE-agent) hits 65% on SWE-bench Verified in ~100 lines (swebench.com, Jul 2025).

WHAT LOOP ENGINEERING IS

The loop: plan -> act -> observe -> verify -> correct --
with a verification gate and a stop condition.



NOUNS vs VERBS

The harness is the nouns (the rig + its sensors). The loop is the verbs (the act -> observe -> verify -> retry cycle).

HARNESS = THE RIG

Nouns

the assembled parts and sensors

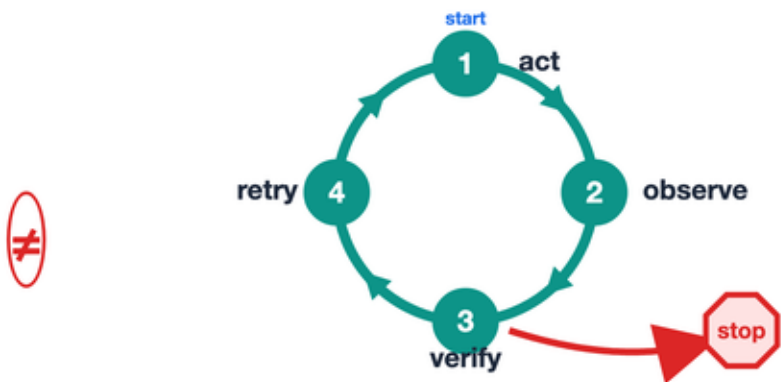


"Everything except the model." What tools and sensors does the agent have?

LOOP = THE CYCLE

Verbs

the cycle and the stop

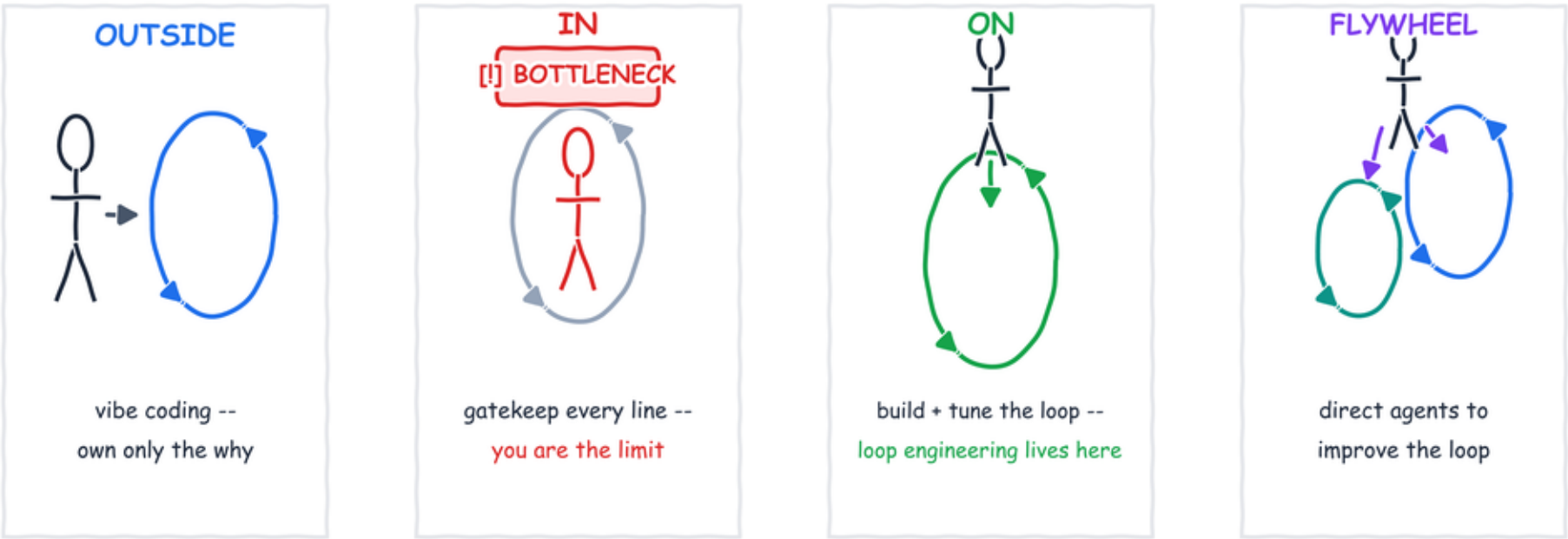


The evaluator-optimizer cycle. How does it iterate, and what makes it stop?

WHO SITS WHERE

Outside, in, on the loop, and the flywheel. 'In the loop' is the trap -- you become the bottleneck.

Who sits where: humans outside, in, on the loop -- and the flywheel

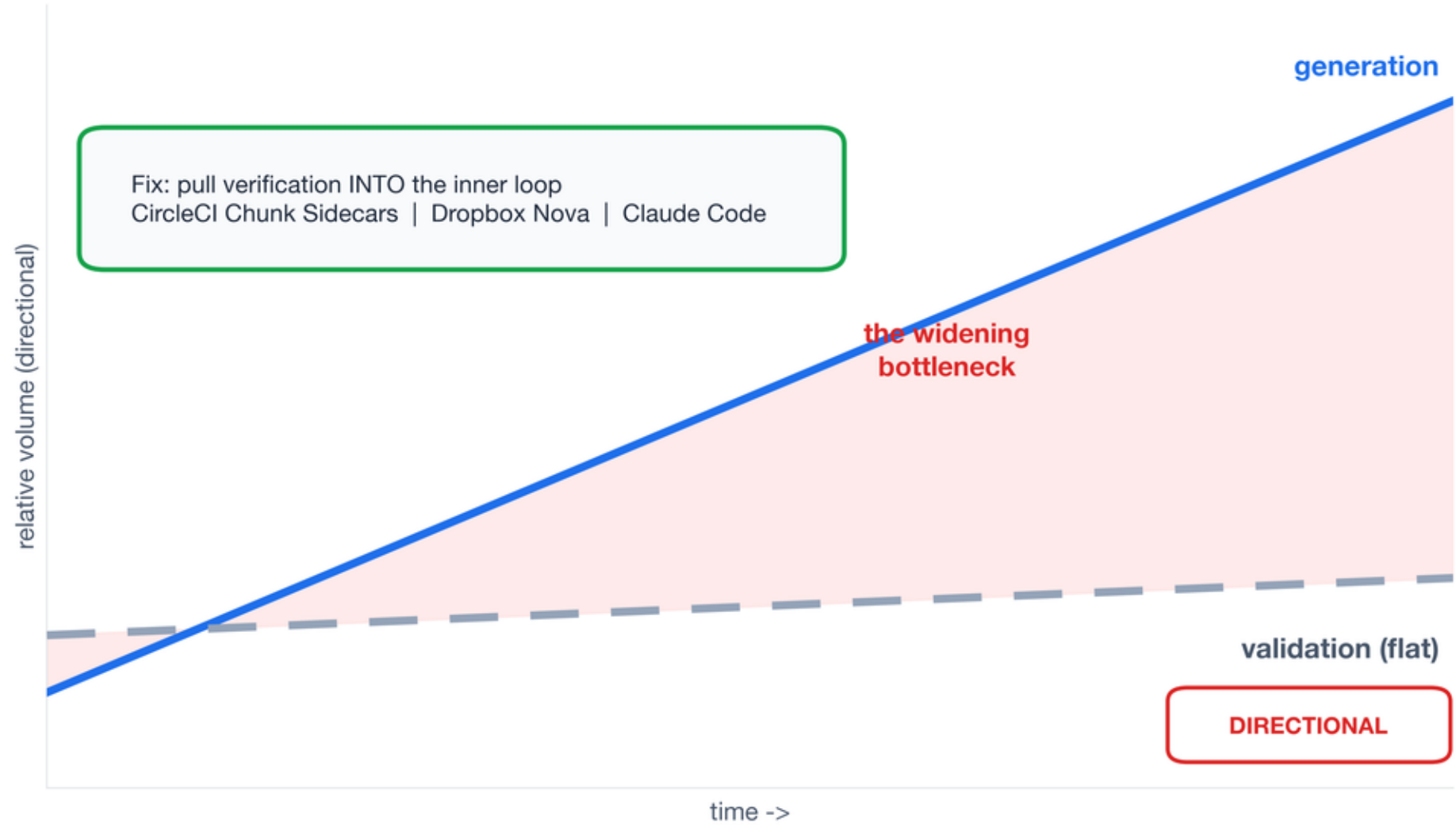


Pivot: IN = fix the output (you edit). ON = fix the loop that produced it (you engineer). Source: Kief Morris, Mar 2026.

WHY NOW

Generation throughput rises while validation stays flat. The fix: pull verification into the inner loop.

Why now: validation, not generation, is the bottleneck



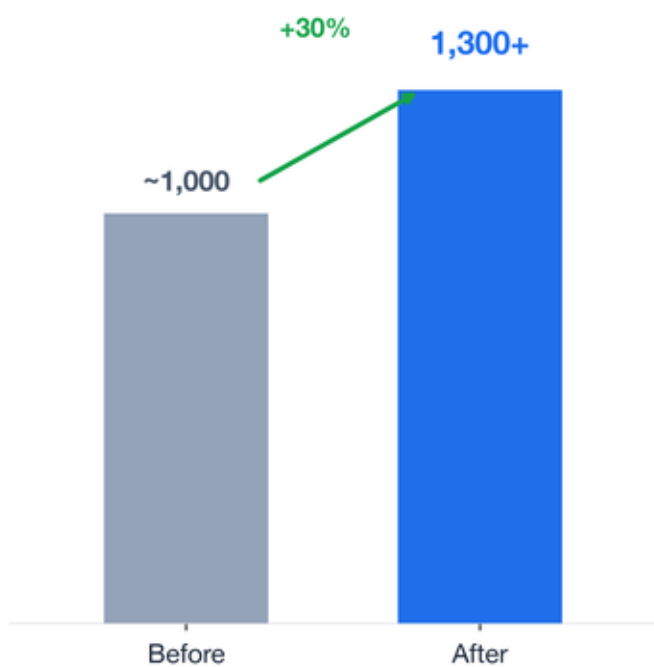
Schematic / directional -- CircleCI's claim is qualitative; axes carry no measured values. Source: CircleCI via InfoQ, Jun 2026.

PROOF AT SCALE

Stripe: ~1,000 -> 1,300+ PRs/week with zero human-written code. SWE-bench Verified: 12.47% -> 76.8% on a fixed harness.

Proof at scale: Stripe Minions and the SWE-bench trajectory

Stripe Minions: PRs / week



Zero human-written code (all human-reviewed) | underpins \$1T+ annual payment volume

Source: InfoQ -> Stripe, Mar 2026

SWE-bench Verified: ~6x in two years



Source: swebench.com, Feb 2026 (v2.0.0). Pricing 'still very subsidized' (Bockeler, Jun 2026).

YOUR FIRST LOOP

A clear goal, the tools to iterate, one machine-checkable feedback signal, and a stop condition.

Your first loop: ship one this week

Take one task where you babysit the agent line-by-line, and give it four things.

1

A clear goal with a success criterion

2

The tools to iterate -- CLI, test runner, linter

3

One machine-checkable feedback signal

4

A stop condition -- max iterations or a definition of done

Source: Anthropic evaluator-optimizer + stop condition; this run's own loop.

BUILD IT YOURSELF

Run a verify->correct loop, build it on a managed runtime, then platform-engineer it with gates.

BUILD IT YOURSELF

Three projects to try this week

Each turns a concept from the post into something you can run — laddering from an afternoon to a weekend.

BEGINNER

Project 1

Run a verify→correct loop in an agent

♦ agent mode · Copilot, Aider, or Claude Code

SUCCESS SIGNAL
your tests exit 0 on an edit the agent made, while you wrote no code

TIME
~90 min

INTERMEDIATE

Project 2

Build the loop on a managed runtime

♦ Foundry Agent Service · or Anthropic agent patterns

SUCCESS SIGNAL
task resolved within your iteration cap; clean exit at the cap

TIME
half a day

ADVANCED

Project 3

Platform-engineer the loop with gates

♦ git-ape · hve-core · mini-swe-agent

SUCCESS SIGNAL
a tightened gate changes the outcome — you fixed the producer

TIME
a weekend

start here

more autonomy →

Each project lists tools you can pick from — choose any. Start with Project 1, then climb.

Which step are you standing on?

Word, context, rig, or loop? In 2026 the unit you own is the loop. Don't just read it -- close one this week.

READ THE FULL ARC

sendtoshailsh.github.io/blog/loop-engineering-ai-native-development.html